Flexible Packaging Costing — Research & Calculator Plan (12 Nov 2026)

How PrintSahaj can cost 2-layer, 3-layer and 4-layer flexible-packaging jobs, including foil, LD, MET, nylon and paper, for every common pouch type, with correct formulas and owner-entered rates.

Research completed **12 Nov 2026 (Asia/Calcutta)**. Densities, microns, waste ranges, cylinder rates and GST are sourced. Every rupee figure in the worked example is a labelled Example, not a market rate.

Three labels used throughout this document

- **Sourced data:** densities, microns, GSM, waste ranges, cylinder rates and GST come from named sources opened on 12 Nov 2026. Nothing in those tables was invented.
- **Example rupees:** every Rs/kg, Rs/pouch and conversion charge in the worked example is a labelled illustration so the arithmetic can be checked.
- **Owner rates:** the calculator gives formulas, a film library and sensible density and coat-weight defaults. Film rates, conversion, wastage, zipper and margin always belong to the owner.

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1. How we handle every structure and every pouch type

Flexible costing looks messy because there are many films and many pouch shapes. The clean way is to split the problem into **two independent inputs**.

1. **The structure** (2-layer, 3-layer, 4-layer, foil, LD, MET, nylon, paper). This only decides laminate GSM and material cost per square metre. The user builds the stack layer by layer: each film is micron times density, paper is entered as mill GSM, each adhesive bond is a dry GSM, and ink is a dry GSM. Two-layer has one bond, three-layer has two, four-layer has three. Foil or MET is just another library item, not a special calculator.

2. **The pouch type** (3-side seal, center seal, standy, side gusset, box pouch, rollstock). This only decides how much film area one pouch uses. A stand-up pouch uses two faces plus an unfolded bottom gusset. A center-seal pillow uses a tube width (two faces plus

back seal). Zipper, spout and valve are not film layers — they are a rupee (and optional gram) add-on per piece.

So the same PET / MET PET / PE structure can cost a 3-side-seal sachet or a zipper standy. The same standy size can be run on a 2-layer or a 4-layer foil retort laminate. Rates stay in the owner's rate book. We do not need a separate tool for every combination — we need **one structure builder, one pouch-geometry picker, and one costing sheet**.

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2. Material data — density, microns, ink and adhesive

Core formula: GSM = micron × density (g/cm³). Yield (m²/kg) = 1,000 ÷ GSM. For a laminate, add every layer and every retained dry coating. Prefer measured GSM from a weighed sample or the supplier TDS when it is available.

Total GSM = Σ (micron × density) + Σ (dry adhesive GSM) + dry ink GSM + other dry coatings

Film densities and typical microns

Film	Default density	Typical microns	Example GSM	Notes and source
PET / BOPET	1.40 g/cm³ (range 1.39–1.41)	8, 10, 12, 15, 19, 23, 36, 50 (print layer often 12 or 23)	12 µm × 1.40 = 16.80 GSM	Indian Cosmo CF-B0-CPF lists density 1.40 g/cc and these exact GSMs. Source: Cosmo Films CF-B0-CPF TDS.
BOPP (plain, standard)	0.905 g/cm³ (range 0.90–0.91)	12, 15, 18, 20, 25, 30, 35, 40, 50 (snacks often 15–25)	20 µm × 0.905 = 18.10 GSM	Planning default 0.905. Matte or cavitated BOPP can be much lighter — a Cosmo matte wet-lamination grade lists 0.84 g/cc. Always use the grade TDS

Film	Default density	Typical microns	Example GSM	Notes and source
				or measured GSM. Sources: RollSpec materials; Balaji International 0.90–0.91; Cosmo PCT-2(M) matte 0.84.
MET BOPP / MET PET	Same as the base film (density unchanged)	MET PET often 12 or 23; MET BOPP often 15–20	Base GSM + about 0.5–1.0 GSM aluminium deposit	Metallisation is a thin aluminium layer on the same PET or BOPP. Prefer measured GSM from the metalliser TDS. Source: CloudFilm density guide.
CPP (cast PP)	0.90 g/cm³ (range 0.89–0.91)	20–80 (retort sealant often 60–80)	60 µm × 0.90 = 54.00 GSM	Used as a heat-seal layer, including retort (higher melt point than PE). Source: RollSpec materials.
LDPE	0.922 g/cm³ (range 0.915–0.928)	25–150 (pouch sealant often 40–100)	70 µm × 0.922 = 64.54 GSM	Blown PE is often a LD + LLD blend. Once a roll is in the factory, use measured GSM. RollSpec notes blown PE can run a few

Film	Default density	Typical microns	Example GSM	Notes and source
				percent heavy. Source: RollSpec materials (nominal 0.922).
LLDPE	0.92 g/cm³ (about 0.915–0.930)	Similar to LDPE; often a co-ex sealant	23 µm × 0.92 = 21.2 GSM (Balaji stretch-film example)	Indian converter Balaji uses 0.92 for cast LLDPE. Treat LD and LLD as separate library items with editable density. Source: Balaji International, Micron to GSM.
HDPE	0.955 g/cm³ (range 0.945–0.965)	10–70 (often a stiff co-ex layer)	30 µm × 0.955 = 28.65 GSM	Heavier than LD at the same micron, so it gives less area per kg. Source: RollSpec materials.
Aluminium foil	2.70 g/cm³ (range 2.69–2.71)	6.35, 7, 9, 12, 15, 20, 25 (India food/pharma often 6–20; 9 is very common)	9 µm × 2.70 = 24.30 GSM	Foil is much heavier than plastic at the same micron. This GSM is metal only — no lacquer or adhesive. Soft temper 8011 / 1235 is the usual packaging foil. Source: RollSpec materials.

Film	Default density	Typical microns	Example GSM	Notes and source
Nylon / BOPA	1.15 g/cm³ (range 1.14–1.16)	12, 15, 20, 25	15 µm = 17.3 GSM (Jindal TDS)	Jindal Nylon Films FILMON BXB (January 2023): density 1.15 g/cc; unit weights 13.8 / 17.3 / 23.0 / 28.8 GSM at 12 / 15 / 20 / 25 µm. Used for puncture and gas barrier, including retort.
Paper / kraft	Do not use density — enter mill GSM	Not used for costing. Paper is bought and specified in GSM.	Tea / ORS / pharma sachets typically 40–70 GSM	Never estimate paper GSM from a generic density (Nalyquo). Source: GT Pack Industries, 30 Mar 2026; Nalyquo laminate GSM notes.
EVOH	1.14 g/cm³ for one verified grade. About 1.12–1.19, depends on ethylene content.	Usually a thin co-ex core, about 3–8 µm	Kuraray EVAL E105B (44 mol% ethylene): 1.14 g/cm³	EVOH density falls as ethylene content rises. Put the grade TDS density in the library. Sources: Kuraray EVAL E105B TDS; CloudFilm range.

Ink and adhesive — dry GSM only

Only the dry mass that stays in the laminate belongs in GSM and in material cost. For solvent-based adhesive, wet kg = dry kg ÷ (solids %). For solventless, solids are 100%, so wet = dry. A 3-layer job has two bonds; a 4-layer job has three. Do not add wet coat weight into laminate GSM — RollSpec shows that this overstates a foil laminate by a large margin.

Item	**Typical dry GSM**	**Source**
Solvent-based dry adhesive (India)	2.0–2.8 GSM dry per bond (normal film/foil); 3.0–3.5 GSM dry on paper or board	Cytech Coatings, Gujarat — HILAM GP-50 application guide
Solvent-based calculator defaults	2.5 GSM dry for one bond (2-ply). 5.0 GSM total for two bonds on a 3-ply (about 2.5 each).	GSMCalculator 2-ply and 3-ply tools
Solventless PU adhesive	Light 1.2–1.8; medium 1.8–2.2; heavy-duty / retort 2.2–2.8 GSM (100% solids, wet = dry)	Toplusse solventless process guide
Gravure ink laydown (starting ranges)	Shallow 1.0–1.2; standard 1.2–1.8; deep / heavy 1.8–2.5 GSM per colour, then × coverage %	GSMCalculator ink tool. Their FAQ also says “wet” — treat as a starting range and prefer measured dry GSM.
Flexo ink laydown (starting range)	0.8–1.5 GSM (same tool). Alliance Printech’s comparison tool defaults are 0.5 (flexo) and 0.4 (gravure) g/m² — those are that tool’s example inputs, not a survey.	GSMCalculator; Alliance Printech flexo vs gravure
Printed-job planning example	RollSpec snack-laminate example uses 1.5 GSM average dry ink on the printed area, plus 1.8 GSM dry solventless adhesive.	RollSpec, How to Calculate Laminate GSM

Ink consumption on a job is **Print area × coverage % × laydown GSM ÷ 1,000**, per colour (GSMCalculator). Coverage starting bands from the same tool: simple text 20–35%, mixed design 40–60%, heavy graphics 60–80%, solid flood 90–100%.

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3. Common Indian structures by end use

These are structures that Indian film makers and converters actually publish. Microns are typical or stated specs, not a national standard. The PrintSahaj calculator should ship with these as presets the owner can edit.

End use	Common structure	Typical / published microns	Source
Biscuits, candy, overwrap	BOPP / BOPP, BOPP / MET BOPP, BOPP / CPP, BOPP / MET CPP	Outer BOPP often 15–20 µm; sealant CPP or BOPP 15–25 µm	Cosmo typical laminates (Technokraft sheet); Kanodia
Snacks, namkeen, cereals (2-layer)	BOPP / PE or PET / PE	BOPP 18–25 or PET 12; PE 40–70	Kanodia laminated-films table; GSMCalculator 2-ply presets
Potato chips, namkeen, gas-flushed snacks (3-layer)	PET / MET PET / PE; PET / extrusion-PE / MET PET / PE; BOPP / MET PET / PE	PET 12 / MET PET 12 / PE 50 is a common calculator preset. Wahren lists 18 BOPP / 12 MET PET / 90 LDPE natural as a real Indian spec.	Cosmo typical laminates; Wahren, Noida
Tea	PET / PE, or PET / MET PET / PE	PET 12; PE as required	Cosmo typical laminates
Instant coffee, spices, export dry food, pharma sachets	PET / Alu foil / PE, or PET / extrusion-PE / Alu foil / PE	GT Pack (Greater Noida, 30 Mar 2026): PET 12 µm / Alu 6–20 µm / PE 60–100 µm. Also	GT Pack Industries; Cosmo; MD PET (2026)

End use	Common structure	Typical / published microns	Source
		BOPP 20 / Alu 9–12 / PE 60–80.	
Pharma / ORS / tea sachets with paper outer	Paper / Alu foil / PE	Paper 40–70 GSM / Alu 9–12 µm / PE 60–80 µm (GT Pack). Wahren examples include 12 µm PET / 17 GSM white PE / 9 µm foil / 35 GSM PE, and heavier 23 PET / 20 foil constructions.	GT Pack; Wahren specifications
Edible-oil sachets	PET / foil / PE	Same foil-laminate family as coffee / pharma	MD PET Packaging Group, Oct 2026
Retort, non-microwaveable (India)	PET / Alu / Nylon (BON) / CPP	Floeter India (Manesar): 12 µm PET / 9 µm Alu / 15 µm BON / 70 µm CPP. Floeter states these pouches can withstand retort up to 125 °C for 30 minutes.	Floeter India, Features and Formats
Retort, microwaveable (no foil)	AlOx PET / Nylon (BON) / CPP	Floeter: 12 µm AlOx PET / 15 µm BON / 70 µm CPP	Floeter India
Retort alternative (foil + CPP, 3-layer)	PET / Alu foil / CPP	GT Pack: PET 12 / Alu 9–12 / CPP 60–80, for 121 °C retort. CPP replaces PE so the seal survives sterilisation.	GT Pack Industries, 30 Mar 2026
Meat, cheese, vacuum, puncture-heavy	Nylon / PE, or PET / Nylon / PE	Nylon 12–15; PE 60. GSMCalculator 3-ply	Kanodia; GSMCalculator 3-ply

End use	Common structure	Typical / published microns	Source
		preset: PET 12 / Nylon 15 / PE 60.	
Atta, rice, heavy dry packs	PET / PE, PE / PE, PET / MET / PE	Thicker PE for drop strength	Cosmo typical laminates

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4. Film-area formulas for each pouch type

First rule, from RollSpec and Aprint: use outer material dimensions that already include the seal lands. If the customer gives only the inside cavity, add the seal widths first. Typical seal width is 5–10 mm (Aprint). Retort seals are often 8–10 mm (RollSpec). Do not add seals twice.

W, H and G are in millimetres. Area A is in square metres. Film grams = A × laminate GSM. Zipper, spout and valve are added after this, as a per-piece cost and optional grams — they are not in the film area (RollSpec, Nalyquo).

3-side seal (3SS / sachet) — two panels, seals already in W and H

A = 2 × W × H / 1,000,000

Front and back. A pouch made from one folded web still has two panels, but one long edge is a fold rather than a seal (RollSpec). If the filler needs an inside size, outside W = inside width + two side seals, and outside H = inside height + bottom seal (and top seal if it is already on the pre-made pouch). Aprint: W and H are outside dimensions, seals included.

Center seal / pillow (VFFS) — web width and cut-off

Web = 2 × face width + total back-seal allowance + total edge trim

A = web × repeat / 1,000,000, with repeat = pouch length including both cross seals.

Source for the web-width formula: PackagingCalculator.org. Aprint: for rollstock, web width is the tube perimeter plus the longitudinal seal, and repeat is pouch length. Fin seal adds the full seal width; a lap seal adds less because the edges overlap — let the owner pick fin or lap.

Stand-up / standy / doypack (bottom gusset) — two faces plus unfolded bottom strip (RollSpec)

A = W × (2H + G) / 1,000,000

G is the full unfolded bottom depth (40 + 40 folded is entered as 80, not 40). The web the pouch machine consumes is about 2H + G; the repeat is about W. Slit width = 2H + G + 2 × edge trim. Written as W × H + G, for example 160 × 240 + 90 mm (Aprint sizing guide). RollSpec worked example: 150 × 220 mm with a 35+35 mm folded gusset (G = 70) gives A = 0.0765 m². This formula is not for side-gusset, quad-seal or shaped pouches.

Side-gusset bag — planning approximation, from RollSpec's perimeter note

A ≈ H × (2W + 2G) / 1,000,000

W is the front-panel width with gussets folded; G is the opened-out depth per side. RollSpec: two side gussets add to the width perimeter, not to the height term. Their example: a 150 mm wide, 220 mm high bag with 70 mm side gussets adds 140 mm to a 300 mm (2 × 150) width perimeter. Aprint writes this format as W × H + G per side. This is a rectangular planning model. Shaped bottoms and handles need a measured dieline.

Flat-bottom, box pouch and quad seal

No simple published area formula was found that is safe to hard-code. RollSpec says these formats have four vertical seals and separate side panels, so both the perimeter and the seal-land count change, and their standard calculator does not cover them. Aprint measures a flat-bottom pouch as front W × H plus a side gusset G, with a formed base of W × G.

Recommendation: let the owner enter a measured dieline area (cm²). Optionally offer the side-gusset formula as a starting estimate, clearly labelled as incomplete for K-seals and formed bottoms. Do not invent an extra percentage.

Other formats and add-ons

- **Rollstock:** cost by running metre and by kg. Area = web width × length. The customer's FFS machine forms the pouch; we can still show an optional implied pouch using their cut-off and tube width (Aprint rollstock notes).
- **Wicketed bags:** treat as 3-side seal plus an owner-entered lip height above the opening, and a wicket-hole allowance. No reliable published lip formula was found; the lip must be an input.
- **Zipper:** length is about pouch width (plus a small trim if the plant adds it). Cost is Rs per metre or Rs per piece, owner-entered. Not part of laminate GSM.

- **Spout / cap / coffee valve:** per-piece bought-out cost, owner-entered. May need a small extra film patch — let that be an optional cm² field.

Layflat and web width

Keep the two directions straight. On a typical single-web stand-up pouch machine, the front, gusset and back are cut across the web, so the slit width is 2H + G plus trim, and the pouch width W is the print repeat along the web (RollSpec). Lane count decides the master printed width: two 570 mm lanes plus 10 mm each side need a 1,160 mm master, not a 1,000 mm roll with huge trim. The gravure cylinder circumference must be a whole multiple of the repeat. Default trim in examples is often 10 mm per edge (RollSpec). This is an input, not a constant. Inter-pouch gaps, if any, add to the repeat.

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5. Costing method, wastage, conversion, cylinders and GST

From layers to a quote

1. **Build laminate GSM.** Films + dry adhesive at each bond + dry ink.
2. **Weighted material cost.** Cost per m² = sum (layer GSM / 1,000 × that layer's Rs/kg). Then Rs/kg of laminate = cost per m² ÷ (total GSM / 1,000).
3. **Pouch film grams.** Developed area × GSM. Pieces per kg = 1,000 ÷ grams.
4. **Add layout trim.** Slit width, not a waste %. This is real purchased film (RollSpec).
5. **Add process waste, last machine first.** Input = good output ÷ (1 − run waste) + setup metres. Do not add trim again inside this percentage.
6. **Add conversion, tooling, hardware, margin and GST.** Conversion is owner rates. Show first-order vs repeat (cylinder on or off). Show GST separately.

A refinement for serious jobs: the printed outer film enters at the press, but the sealant film enters only at lamination, so the two webs do not need the same length (RollSpec). Version 1 of the website calculator can use one combined kg figure; the quote engine in Flexora should split the webs.

Wastage — published ranges, not our invention

Stage	Published figure	How to treat it	Source
Printing	2–4% typical (GSMCalculator FAQ). RollSpec worked example uses	Owner-editable. Setup metres matter more on short jobs.	GSMCalculator pouch costing FAQ; RollSpec order-quantity guide

Stage	Published figure	How to treat it	Source
	3% run waste + 800 m setup.		
Lamination	1–3% typical (GSMCalculator). RollSpec example: 2% + 400 m (includes curing samples).	Same — percentage plus optional setup metres.	Same two sources
Slitting	1–2% typical (GSMCalculator). RollSpec example: 1% + 100 m.	Edge trim from a wrong master width is not this percentage — it is a layout error.	Same two sources
Pouching	RollSpec example uses 4% run waste (reel changes included). GSMCalculator's FAQ does not give a pouching percentage.	Label RollSpec's 4% as an illustrative input, which is how RollSpec itself describes it — not an industry benchmark.	RollSpec: illustrative inputs, not industry benchmarks
Combined	GSMCalculator: a combined wastage of 3–5% is typical for standard jobs.	Acceptable only as a simple-mode default. Better: stage-wise, last machine first, and compound (input = output ÷ 0.96, not +4% of output).	GSMCalculator

Conversion charges per kg — no reliable published Indian schedule

No reliable published Indian schedule of conversion charges per kg (printing, lamination, slitting, pouching) was found. Trade-portal finished-pouch Rs/kg figures mix material and conversion and are not usable as defaults. Alliance Printech's Indian pouch-cost tool builds conversion from machine time, speed, electricity and operators, not from a published Rs/kg list.

PrintSahaj should give empty, owner-editable lines: printing (gravure or flexo), lamination, slitting, pouching — each as Rs/kg or Rs/1,000, plus optional Rs/hour machine time. Example numbers in demos must wear an **Example** badge. After a few jobs, Flexora can replace these with the plant's own history.

Cylinder and plate cost (India)

Item	Published or tool figure	Date / notes	Source
Gravure cylinder, Kanodia Gravure (Delhi)	Rs 2.10 / cm² when circumference is below 750 mm; Rs 2.30 / cm² above 750 mm. Cost = length (cm) × circumference (cm) × rate, then × number of colours.	Published on Kanodia's own price estimator. Page retrieved 12 Nov 2026. Their website formula block is a little garbled (divided by 100); the stated method is area in cm² × Rs/cm², with length and circumference entered in cm.	kanodiagravure.com/cylinder-costs
Alliance Printech tool defaults (Faridabad)	Gravure engraving default Rs 2.40 / cm². Flexo plate default Rs 1.50 / cm².	These are pre-filled inputs in a free comparison tool, retrieved 12 Nov 2026 — useful starting defaults, not a market survey. The page also explains why gravure tooling dominates short-run cost, and why a break-even quantity exists versus flexo.	allianceprintech.com/resources/flexo-vs-gravure

Cylinder cost is a fixed job cost, spread over the order (and over reprints if the same cylinders are reused). On a short first order it can be more than the film. The calculator must show **first-order** and **repeat-order (no cylinder)** side by side.

GST on flexible packaging in India

For the usual plastic films, laminated plastic films, and plastic pouches, the rate in the current GST schedules is **18%** (9% CGST + 9% SGST, or 18% IGST).

- **HSN 3920** (plain plastic film, not laminated): 18%, Notification 9/2025-Central Tax (Rate), Schedule II, S. No. 121.
- **HSN 3921** (other plates, sheets and film, including flexible laminated / metallised tariff items): 18%, same notification, S. No. 122.
- **HSN 3923** (plastic articles for packing of goods, including bags and pouches): 18%, S. No. 124.

Compiled by FundRaksha from CBIC Notification 9/2025-CT(R), as of 21 Oct 2026, following the 22 Sep 2025 rate reform. Paper sacks and some biodegradable bags can be 5% under a chapter-wide entry — do not assume 18% for every paper-heavy pack. Foil-heavy constructions can raise a classification question (Chapter 76 versus 39). The quote PDF should default to 18% and tell the owner to confirm HSN with their CA. **This document is not tax advice.**

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6. Existing calculators — India first, then international

Each URL below returned HTTP 200 on 12 Nov 2026, except where noted. “Best to copy” means ideas, not their code or paid content.

India

Tool	What it does well	Gaps	Copy for PrintSahaj?
GSMCalculator Packtools. Free web. Currency examples in AED, but the formulas are general. 2-ply, 3-ply, 4-ply, pouch weight, pouch costing, ink, cylinder, roll tools. All listed pages loaded (200).	Layer-by-layer GSM and cost per m²; adhesive defaults (2.5 / 5.0 GSM); common Indian-style structure presets; ink coverage bands; wastage FAQ with percentage ranges; cylinder amortisation.	The pouch-weight page describes area as width × height — that can under-count a two-panel pouch. Stand-up gusset, side gusset and zipper are weak or missing. Not a saved rate-book for an Indian owner.	Yes — structure presets, adhesive defaults, ink coverage, 2/3/4-ply layout. Do not copy a one-panel area formula.
Agarwal Packaging System, Chennai. Free pouch and film cost calculator. Loaded (200).	Indian, simple, and in the right language of the trade: pillow / centre-seal or 3-side seal, layers with micron and Rs/kg, printing and	Only two pouch types. No standy, zipper or foil-specific guidance on the public page. Built to sell packing	Yes — the simple input order and the selling-price-with-mar gin output.

Tool	What it does well	Gaps	Copy for PrintSahaj?
	lamination, wastage, extra per pouch, margin, cost per pouch and per 1,000.	machines, not to be a full converter tool.	
Alliance Printech pouch cost and Flexo vs gravure. Faridabad press maker. Both loaded (200).	Costs a job from film, plates or cylinders, ink, press time and converting time. The comparison tool shows a break-even quantity between flexo and gravure — something PrintSahaj can offer as a teaching feature.	Full breakdown is emailed; it is a press-sales tool. Defaults (Rs 2.40/cm², Rs 1.50/cm²) are examples. Not a structure library.	Yes — first-order vs repeat, flexo vs gravure break-even, tooling spread.
Kanodia Gravure cylinder estimator, Delhi. Loaded (200).	A real Indian published Rs/cm² rate, with a circumference split at 750 mm.	Cylinders only, not pouch costing.	Yes — use as a labelled default the owner can overwrite.
Wiserobe Flexible Packaging Toolkit. Paid Excel, Rs 149 + add-ons. Sales page loaded (200). Seller: AVSVishal Media, Jaipur.	Closest Indian product to a full converter workbook: layflat by pouch style, micron-to-GSM, structure cost, ink and adhesive, cylinder share, printing / lamination / slitting / making, stage wastage, zipper and spout, cost per 1,000, slab rates, quotation, profit. Rates are the user's (yellow input cells).	Paid download, not a free web magnet. Job-card / CRM / stock kits are add-ons — those belong in Flexora, not on the free PrintSahaj site. Earlier review also flagged checkout dark patterns; we copy costing ideas only.	Yes — the costing workflow and owner-rate design. No for job cards and CRM on the free site.

Tool	What it does well	Gaps	Copy for PrintSahaj?
Gedgetsworld toolkit. Paid pack, Agra seller. Page loaded (200).	Similar pitch: costing plus production templates.	Earlier review found fewer real calculators, generic Excel templates, and a refund policy that contradicts the 30-day guarantee. Weaker model to copy.	Little — do not follow this product design.

International

Tool	What it does well	Gaps	Copy for PrintSahaj?
RollSpec tools and pouch-weight guide. Engineering calculators and long worked guides. Loaded (200). Last technical review on the tools page: 5 Sep 2026.	The most careful free treatment found: correct stand-up formula, seal-land warning, trim versus process waste, waste worked backwards with setup metres, different entry points for PET and PE webs, adhesive dry vs wet, and explicit illustrative-not-a-benchmark labels.	Not an Indian rate book. Side-gusset and box pouch are explained in words, not fully calculated. No quote PDF or GST.	Yes — this should be the engineering backbone of our formulas and warnings.
Nalyquo laminate GSM, pouch-weight, and the full flexible suite. Large free suite. Loaded (200). Pages dated Oct 2026.	Up to six layers; dry adhesive, ink and coating as separate lines; paper must be real GSM; pouch weight can use production web (width × repeat ÷ lanes × GSM) or a flat estimate;	Flat estimate is not a gusset model. No Indian GST or cylinder Rs/cm². Many tools are purchasing-oriented.	Yes — web-vs-flat methods, assumption footnotes, adhesive solids, price per kg vs per m².

Tool	What it does well	Gaps	Copy for PrintSahaj?
	assumptions are written on the page.		
Microflex Film, US. 15 free estimators. Loaded (200).	Excellent format picker, barrier selector and pouch-size intuition, with diagrams. Good teaching UX.	Not a converter costing tool. No GSM stack, no Rs/kg, no Indian structures.	Yes — pouch-type diagrams and a simple which-structure-for-this-product helper.
PackagingCalculator.org web width. Loaded (200). Also has pouch capacity, roll length, repeat tools.	Clear center-fold web-width formula with seal and trim as totals, not per-side traps.	Not a full cost sheet.	Yes — the web-width inputs and wording.
Cerm, Kiwiplan, PCMC, ePac, Flexible Packaging Association	Cerm is a paid print / flex-pack MIS. Kiwiplan is mainly corrugate MES. PCMC builds pouch and bag machines. ePac is a converter network. FPA is a trade association. All are relevant names in the industry.	No public costing calculator was found on these sites during this research. They should not be listed to a user as free tools they can open today.	No public formula to copy. Flexora, later, can learn from what a MIS tracks (job ticket, actual vs estimate) — not from a public calculator.

Bottom line: no free tool already does Indian 2/3/4-layer structure + every pouch type + zipper + owner-saved rates + GST + quote PDF in one place. That is the gap PrintSahaj should fill. The best engineering source is RollSpec; the best Indian workflow inspiration is Wiserobe's costing sheet and Agarwal's simplicity; the best published cylinder rupees are Kanodia's.

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7. Recommended calculator design for printsahaj.com

Six free tools, one shared rate book. Serious production control stays in Flexora.

Screen 1 — Structure builder

Preset Indian structures (chips, coffee, pharma, retort, biscuits), fully editable. Layer picker: PET, BOPP, MET PET, MET BOPP, CPP, LDPE, LLDPE, HDPE, Alu foil, BOPA, paper (GSM), EVOH, custom. Each film layer: micron + density (library default, owner can change) or a measured GSM override. An adhesive row appears automatically between every two films. Default dry GSM from Cytech / GSMCalculator; solvent-based or solventless (solids %). Ink: colours, coverage %, dry laydown, optional white underprint. Live result: total GSM, yield m²/kg, Rs/m² and Rs/kg at the owner's film rates.

Screen 2 — Pouch type and size

Picker with a small diagram: 3SS, center-seal / pillow, standy (bottom gusset), side gusset, flat-bottom / quad (dieline area), rollstock, wicketed. W, H, G. Toggle: these are outer sizes (default) or these are inside, add seals. Seal width default 8 mm, editable (5–10 mm sourced range). G for standy is unfolded depth, with a helper: if your gusset is 40+40 folded, type 80. Fin vs lap for center seal. Lip height for wicketed. Dieline cm² for box / quad. Zipper, spout, valve: on/off, Rs/piece (and optional grams). Show developed area, slit width, suggested repeat, and a warning if cylinder circumference is not a multiple of the repeat.

Screen 3 — Quantity, waste, rates

Order quantity, lanes, edge trim (default 10 mm/side, labelled as a common example from RollSpec), optional setup metres. Waste by stage, with sourced starting ranges and a clear not-a-benchmark note. Simple mode: one combined 3–5% default from GSMCalculator. Rate book (login to save): Rs/kg per film, adhesive, ink; conversion lines empty or last-saved; cylinder Rs 2.10 / Rs 2.30 default from Kanodia, flexo plate Rs 1.50 default from Alliance, both labelled. Margin %. GST 18% default, editable. All defaults that are money must show an **Example** or starting-default-change-me tag until the owner saves their own.

Screen 4 — Outputs and the lead gate

Grams per pouch (film only, and film + zipper), pieces/kg, kg of each web to buy, cost per pouch, per 1,000, per kg. First-order vs repeat (cylinder / plate on or off). Slab rates at 10k, 25k, 50k, 100k if they type a margin. Breakdown: material, conversion, tooling share, hardware, waste, margin. GST shown separately. Free on-screen result for tools 1–5 (structure, pouch cost, roll/label cost, cylinder spread, ink and adhesive). Lead form before Quote PDF: name, WhatsApp, company, city. PDF with PrintSahaj logo, GST split, and a Flexora demo line. Saving a permanent rate book also requires a login — that is the second lead moment.

Keep off the free site

Job cards, CRM, film stock, worker attendance and production tracking belong in **Flexora**, offered after the quote PDF or a saved rate book.

8. Worked example — 3-layer standy with zipper

All rupee rates below are examples only, except the cylinder rate. GSM, densities and the area formula are sourced. The Rs/kg and conversion figures are not market prices and were not taken from any supplier quote. They exist so the arithmetic is visible and checkable. An owner must replace every rupee with their own rate.

Inputs

Input	Value	Why this value
Pouch	Stand-up, 160 × 240 + 80 mm, with zipper. Quantity 50,000. Outer sizes, seals already included.	Aprint notation W × H + G. G = 80 mm unfolded (for example 40+40 folded), as RollSpec requires.
Structure	PET 12 / MET PET 12 / LDPE 70, two adhesive bonds, printed.	A common Indian 3-layer snack / coffee barrier style (Cosmo, GSMCalculator, Wahren family). Microns chosen as a typical preset, not a customer's actual TDS.
Densities and coat weights	PET 1.40; MET PET 1.40 + 0.80 GSM metal; LDPE 0.922; adhesive 2.50 dry × 2 bonds; ink 1.50 dry average.	PET: Cosmo CF-B0-CPF. Metal add-on: mid-point of CloudFilm's 0.5–1.0 GSM (labelled assumption). LDPE: RollSpec 0.922. Adhesive: Cytech / GSMCalculator 2.5. Ink: RollSpec snack example 1.5 dry.
Trim and waste	10 mm edge trim each side. Combined process waste 5% of input after trim (simple mode).	10 mm is RollSpec's worked-example trim, used here as an example input. 5% is the top of GSMCalculator's typical combined 3–5%. Setup metres are left out in this simple run; a short job should add them.

Input	Value	Why this value
Example rates (Example — not sourced, except the cylinder rate)	PET Rs 145/kg, MET PET Rs 185/kg, LDPE Rs 125/kg, adhesive Rs 400/kg dry, ink Rs 550/kg dry, conversion Rs 76/kg total (print 28 + lam 18 + slit 8 + pouch 22), zipper Rs 0.90/piece, 6 gravure colours, margin 15%.	Invented for the demonstration. Cylinder rate is the one money exception: Rs 2.10/cm² from Kanodia, because circumference 640 mm is below 750 mm.

Step 1 — Laminate GSM (sourced arithmetic)

Layer	Calculation	GSM
PET 12 µm	12 × 1.40	16.80
MET PET 12 µm + metal	12 × 1.40 + 0.80	17.60
LDPE 70 µm	70 × 0.922	64.54
Dry adhesive, two bonds	2.50 + 2.50	5.00
Dry ink (average)	1.50	1.50
Total laminate	16.80 + 17.60 + 64.54 + 5.00 + 1.50	**105.44 g/m²**

Yield = 1,000 ÷ 105.44 = **9.484 m²/kg**.

Step 2 — Film area and grams (RollSpec formula)

A = 160 × (2 × 240 + 80) / 1,000,000 = 160 × 560 / 1,000,000 = 0.0896 m²

Film grams (blank) = 0.0896 × 105.44 = 9.4474 g. Theoretical pieces/kg = 1,000 / 9.4474 = 105.85. Web the machine consumes = 2H + G = 560 mm. With 10 mm trim each side, slit width = 580 mm. Repeat = 160 mm (no gap in this example).

Purchased area = 160 × 580 / 1,000,000 = 0.0928 m². Grams with trim = 0.0928 × 105.44 = 9.7848 g. Good film for 50,000 pouches = 50,000 × 9.7848 / 1,000 = 489.24 kg.

This 3.45% is layout trim, already purchased. It is not added again as process waste.

Step 3 — Process waste (simple mode). 5% of input, after trim

Input kg = 489.24 ÷ 0.95 = 515.00 kg (489.24 / 0.95 = 514.99, rounded to 515.00). Input area is about 515.00 ÷ 105.44 × 1,000 = 4,884 m².

A fuller plant model would compound 3% print, 2% lam, 1% slit and 4% pouch from the last machine backwards, and would buy a slightly shorter PE web than PET web. That is the Flexora-grade version.

Step 4 — Material and conversion rupees (Example rates)

Line	**Calculation**	**Rs / m²**
PET	16.80 / 1,000 × 145	2.436
MET PET	17.60 / 1,000 × 185	3.256
LDPE	64.54 / 1,000 × 125	8.0675
Adhesive	5.00 / 1,000 × 400	2.000
Ink	1.50 / 1,000 × 550	0.825
Material	sum, then × 1,000 / 105.44 = **Rs 157.29 / kg**	**16.5845**

Cost line for 50,000 pouches	**Calculation**	**Amount**
Material, including trim and 5% waste	515.00 kg × Rs 157.29	Rs 81,004
Conversion (Example)	515.00 × Rs 76	Rs 39,140
Gravure cylinders, first order (6 colours). Circumference 640 mm (4 × 160 mm repeat), face length 600 mm assumed for a 580 mm slit plus margin. Area = 64 cm × 60 cm = 3,840 cm². Rate Rs 2.10 (Kanodia, below 750 mm).	3,840 × 2.10 × 6	Rs 48,384

Cost line for 50,000 pouches	Calculation	Amount
Zipper (Example)	50,000 × Rs 0.90	Rs 45,000
Total factory cost, first order	81,004 + 39,140 + 48,384 + 45,000	**Rs 2,13,528**
Repeat order (no new cylinders)	2,13,528 − 48,384	**Rs 1,65,144**

Step 5 — Per pouch, margin and GST

Output	First order	Repeat order
Cost per pouch	Rs 4.27	Rs 3.30
Cost per 1,000	Rs 4,271	Rs 3,303
Selling price at 15% margin, before GST (Example margin)	Rs 4.91 / pouch	Rs 3.80 / pouch
GST 18% (HSN typically 3923 for pouches)	Rs 0.88	Rs 0.68
Invoice price including GST	Rs 5.79 / pouch	Rs 4.48 / pouch

Headline figures (Example rates, 50,000 pcs):

- First-order cost / pouch: **Rs 4.27**
- Repeat-order cost / pouch: **Rs 3.30**
- Cylinder share / pouch: **Rs 0.97** (why first and repeat must differ)
- Laminate GSM: **105.44** (sourced densities and coat weights)

Rounded to the nearest paisa from Rs 2,13,528 ÷ 50,000 = Rs 4.2706, then × 1.15 = Rs 4.9111, then × 1.18 = Rs 5.7951. Cylinder share on the first order is Rs 0.97 per pouch — that is why a 50,000 first run and a repeat run must not be quoted at the same rate. Zipper weight is not in the 9.78 g film figure. If the plant sells on finished empty-pouch weight, the owner adds a measured zipper mass (weigh 20 finished pouches, as RollSpec recommends).

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9. Sources, opened 12 Nov 2026

Research started with Indian film makers, adhesive makers, converters and GST compilations, then international engineering calculators. HTTP 200 was confirmed with a request on 12 Nov 2026 unless noted. Densities are typical or TDS values for named grades, not purchasing specifications. Replace them with the owner's supplier TDS or a weighed GSM. Conversion Rs/kg and film Rs/kg are not published as a trustworthy Indian schedule in the sources above. GST classification of unusual foil or paper packs should be confirmed with a qualified professional.

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2. Cosmo Films PCT-2(M) matte BOPP TDS — density 0.84 g/cc on a matte grade
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5. Kuraray EVAL E105B TDS — EVOH density 1.14 g/cm³ (HTTP 200)
6. RollSpec film density and yield table (HTTP 200)
7. RollSpec laminate GSM guide
8. RollSpec pouch-weight guide — stand-up formula and seal-land warning
9. RollSpec adhesive consumption — dry vs wet, solids %
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11. CloudFilm density, metallised add-on 0.5–1.0 GSM, EVOH range (HTTP 200)
12. Balaji International, India — micron to GSM (page retrieved; some automated fetches returned 403)
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14. Toplusse solventless coat-weight bands (HTTP 200)
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16. Wahren, Noida — published laminate specifications (HTTP 200)
17. Kanodia Technoplast — structure-by-application table (HTTP 200)
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30. Nalyquo laminate GSM calculator (HTTP 200)
31. Nalyquo pouch weight calculator (HTTP 200)
32. Microflex Film packaging calculators (HTTP 200)
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